Chapter 4

Campbell Biology, AP Edition, 12e

Qinling golden snub-nosed monkeys gathered around a forest stream.

Chapter 4********

Carbon and the Molecular Diversity of Life

Figure 4.1

These Qinling golden snub-nosed monkeys in the mountainous forests of southwest China, like all living organisms, are made up of chemicals based mostly on the element carbon. Of all chemical elements, carbon is unparalleled in its ability to form molecules that are large, complex, and varied, making possible the diversity of organisms that have evolved on Earth.

What makes carbon the basis for all biological molecules?

Carbon can form four bonds and therefore can bond up to four other atoms or groups of atoms. Carbon can bond to other carbons, resulting in carbon skeletons. Carbon also commonly bonds to hydrogen, oxygen, and nitrogen. The properties of a carbon-containing molecule depend on the arrangement of its carbon skeleton and on its chemical groups. The signaling molecule that is shown here, dopamine, has many functions, including promoting mother-infant bonding.

Key Concepts

- **4.1Organic chemistry is key to the origin of life****
- **4.2Carbon atoms can form diverse molecules by bonding to four other atoms****
- **4.3A few chemical groups are key to molecular function****

Advanced placement edition

Carbon is one of the key elements around which living matter is organized and core biological processes have evolved **(BIG IDEA 1)**. It plays a vital role in the capture, use, and storage of energy needed to support life **(BIG IDEA 2)**, and it serves as the backbone for a great diversity of biomolecules that are synthesized and/or degraded as it cycles through the environment **(BIG IDEA 4)**.

Study Tip

Label chemical groups: After you have read through Figure 4.9, look through Chapters 4 and 5 for molecules that have the chemical groups shown in that figure. Circle and label the chemical groups you find, as in the following example:

Estradiol and testosterone have hydroxyl groups, methyl groups, and a carboxyl group.

Get Ready for This Chapter

In this chapter, you will learn how carbon functions as the basis of life's molecules. You will also learn about some common small molecules and chemical groups. Before beginning this chapter, you should be able to do the following:

- Describe in general the structure of atoms and the meaning of atomic number and atomic mass (see Concept 2.2).
- Explain how electrons are distributed in an atom's electron shells and how the electron distribution determines the chemical behavior of an atom (see Concept 2.2, section entitled "Electron Distribution and Chemical Properties").
- Explain what covalent bonds are and how they are formed (see Concept 2.3).
- Explain the bonding behavior of a carbon atom (see Concept 2.3).

The following questions provide a quick check of your basic knowledge in these areas.

1. An atom of iron has the atomic number 26. This means that

it has 52 protons and 52 electrons.

That's incorrect. Try again.

Incorrect. Review the following material: The atoms of a particular element have a unique number of protons in their nuclei, and this number is called the atomic number. Unless otherwise indicated, an atom is neutral in electrical charge, which means that its protons must be balanced by an equal number of electrons. Therefore, since the atomic number tells us the number of protons, it also tells us the number of electrons in an electrically neutral atom.

it has 26 protons and 26 electrons.

That's correct.

Correct! If you had trouble with this question, review the following material: The atoms of a particular element have a unique number of protons in their nuclei, and this number is called the atomic number.

Unless otherwise indicated, an atom is neutral in electrical charge, which means that its protons must be balanced by an equal number of electrons. Therefore, since the atomic number tells us the number of protons, it also tells us the number of electrons in an electrically neutral atom.

it has 13 neutrons, 13 protons, and 13 electrons.

That's incorrect. Try again.

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it has 13 electrons, 26 protons, and 26 neutrons.

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Incorrect. Review the following material: The atoms of a particular element have a unique number of protons in their nuclei, and this number is called the atomic number. Unless otherwise indicated, an atom is neutral in electrical charge, which means that its protons must be balanced by an equal number of electrons. Therefore, since the atomic number tells us the number of protons, it also tells us the number of electrons in an electrically neutral atom.

C: atomic mass 12, atomic number 6. O: atomic mass 16, atomic number 8. H: atomic mass 1, atomic number 1. N: atomic mass 14, atomic number 7. S: atomic mass 32, atomic number 16. P: atomic mass 31, atomic number 15.

2. Based on the electron configuration of the elements in the figure, which would exhibit a chemical behavior most like that of oxygen?

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sulfur
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That's correct.

Correct! If you had trouble with this question, review the following material: Oxygen (O) has two electrons in its first shell and six electrons in its valence shell. Sulfur (S) has two electrons in its first shell, eight electrons in its second shell, and six electrons in its valence shell. Atoms with the same number of electrons in their valence shells exhibit similar chemical behavior. For example, both oxygen and sulfur can form two single covalent bonds with two hydrogen atoms to complete their valence shells.

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carbon
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That's incorrect. Try again.

Incorrect. Review the following material: Oxygen (O) has two electrons in its first shell and six electrons in its valence shell. Sulfur (S) has two electrons in its first shell, eight electrons in its second shell, and six electrons in its valence shell. Atoms with the same number of electrons in their valence shells exhibit similar chemical behavior. For example, both oxygen and sulfur can form two single covalent bonds with two hydrogen atoms to complete their valence shells.

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phosphorus
```

That's incorrect. Try again.

Incorrect. Review the following material: Oxygen (O) has two electrons in its first shell and six electrons in its valence shell. Sulfur (S) has two electrons in its first shell, eight electrons in its second shell, and six electrons in its valence shell. Atoms with the same number of electrons in their valence shells exhibit similar chemical behavior. For example, both oxygen and sulfur can form two single covalent bonds with two hydrogen atoms to complete their valence shells.

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nitrogen
```

That's incorrect. Try again.

Incorrect. Review the following material: Oxygen (O) has two electrons in its first shell and six electrons in its valence shell. Sulfur (S) has two electrons in its first shell, eight electrons in its second shell, and six electrons in its valence shell. Atoms with the same number of electrons in their valence shells exhibit similar chemical behavior. For example, both oxygen and sulfur can form two single covalent bonds with two hydrogen atoms to complete their valence shells.

3. An atom has four electrons in its valence shell. What type(s) of covalent bonds is it capable of forming with oxygen?

```
double bonds only
```

That's incorrect. Try again.

Incorrect. Review the following material: The number of electrons in an atom’s valence shell determines its bonding capacity. The type of covalent bond an atom with four valence electrons is capable of forming with oxygen will be limited by oxygen. Oxygen has six valence electrons, and can form two covalent bonds (one double bond or two single bonds) to complete its valence shell.

```
single and double bonds
```

That's correct.

Correct!If you had trouble with this question, review the following material: The number of electrons in an atom’s valence shell determines its bonding capacity. The type of covalent bond an atom with four valence electrons is capable of forming with oxygen will be limited by oxygen. Oxygen has six valence electrons, and can form two covalent bonds (one double bond or two single bonds) to complete its valence shell.

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```
it cannot form covalent bonds with oxygen
```

That's incorrect. Try again.

Incorrect. Review the following material: The number of electrons in an atom’s valence shell determines its bonding capacity. The type of covalent bond an atom with four valence electrons is capable of forming with oxygen will be limited by oxygen. Oxygen has six valence electrons, and can form two covalent bonds (one double bond or two single bonds) to complete its valence shell.

Figure showing carbon A with a single line connected to carbon B.

4. How many hydrogen atoms can be attached to carbon A?

1

That's incorrect. Try again.

Incorrect. Review the following material: Each carbon atom can form four bonds. Each hydrogen atom can form one bond. Carbon A has one single bond to the adjacent carbon in the diagram, so it can form three more single bonds with three hydrogen atoms.

2

That's incorrect. Try again.

Incorrect. Review the following material: Each carbon atom can form four bonds. Each hydrogen atom can form one bond. Carbon A has one single bond to the adjacent carbon in the diagram, so it can form three more single bonds with three hydrogen atoms.

3

That's correct.

Correct! If you had trouble with this question, review the following material: Each carbon atom can form four bonds. Each hydrogen atom can form one bond. Carbon A has one single bond to the adjacent carbon in the diagram, so it can form three more single bonds with three hydrogen atoms.

4

That's incorrect. Try again.

Incorrect. Review the following material: Each carbon atom can form four bonds. Each hydrogen atom can form one bond. Carbon A has one single bond to the adjacent carbon in the diagram, so it can form three more single bonds with three hydrogen atoms.

Learning Objectives

After studying this chapter, you should be able to do the following:

- 4.1Explain what organic compounds are, and how they might have arisen on Earth.
- 4.2Use examples to demonstrate how carbon's atomic structure results in a wide range of molecular structures.
- 4.3Identify the key chemical groups that affect the function of biological molecules.

Concept 4.1: Organic chemistry is key to the origin of life

For historical reasons, compounds containing carbon are said to be organic, and their study is called **organic chemistry**. Organic compounds range from simple molecules, such as methane (CH_4), to colossal ones, such as proteins, with thousands of atoms.

EVOLUTION In 1953, Stanley Miller, a graduate student of Harold Urey at the University of Chicago, designed an experiment on the abiotic (nonliving) synthesis of organic compounds to investigate the origin of life. Study **Figure 4.2** to learn about his classic experiment. From his results, Miller concluded that complex organic molecules could arise spontaneously under conditions thought at that time to have existed on early Earth. You can work with the data from a related experiment in the **Scientific Skills Exercise**. These experiments support the idea that abiotic synthesis of organic compounds, perhaps near volcanoes, could have been an early stage in the origin of life (see Figure 25.2).

Figure 4.2

Inquiry: Can organic molecules form under conditions estimated to simulate those on the early Earth?

Experiment

In 1953, Stanley Miller set up a closed system to mimic conditions thought at that time to have existed on the early Earth. A flask of water simulated the primeval sea. The water was heated so that some vaporized and moved into a second, higher flask containing the “atmosphere”—a mixture of gases. Sparks were discharged in the synthetic atmosphere to mimic lightning.

Stanley Miller experiment tests whether organic molecules form under conditions that are estimated to simulate early-Earth conditions.

Experiment: in 1953, Stanley Miller set up a closed system to mimic conditions thought at the time to have existed on early Earth. A flask of water simulated the primeval sea. The water was heated so that some vaporized and moved into a second, higher flask containing the atmosphere- a mixture of gasses. Sparks were discharged by electrodes in the synthetic atmosphere to simulate lightning. Pictured is the experimental apparatus consisting of flasks connected by tubes that allow for the movement of water as

vapor when heated or, when cooled, organic molecule containing rain, as well as flasks that allow for the collection of samples for chemical analysis.

Results

Miller identified a variety of organic molecules that are common in organisms. These included simple compounds, such as formaldehyde (CH_2O) and hydrogen cyanide (HCN), and more complex molecules, such as amino acids and long chains of carbon and hydrogen known as hydrocarbons.

Conclusion

Organic molecules, a first step in the origin of life, may have been synthesized abiotically on the early Earth. Although later evidence indicated that the early-Earth atmosphere was different from the "atmosphere" used by Miller in this experiment, recent experiments using the revised list of chemicals also produced organic molecules. (We will explore this hypothesis in more detail in Concept 25.1.)

Data from S. L. Miller, A production of amino acids under possible primitive Earth conditions, *Science* 117:528–529 (1953).

WHAT IF? *If Miller had increased the concentration of* NH_3 *in his experiment, how might the relative amounts of the products* HCN *and* CH_2O *have differed?*

*For suggested answer, see*Appendix A.**

Scientific Skills Exercise

Working with Moles and Molar Ratios

AP Science Practices 5, 6

Could the First Biological Molecules Have Formed Near Volcanoes on Early Earth? In 2007, Jeffrey Bada, a former graduate student of Stanley Miller, discovered some vials of samples that had never been analyzed from an experiment performed by Miller in 1958. In that experiment, Miller used hydrogen sulfide gas (H_2S) as one of the gases in the reactant mixture. Since H_2S is released by volcanoes, the H_2S experiment was designed to mimic conditions near volcanoes on early Earth. In 2011, Bada and colleagues published the results of their analysis of these "lost" samples. In this exercise, you will make calculations using the molar ratios of reactants and products from the H_2S experiment.

Stanley Miller's notes and vials.

A note from Stanley Miller's journal numbered 114 on the top left and dated March 24, 1958 is shown. An experimental run number 22 is also seen. The following is written below:

C H 4: 25.8

C O 2: 8.7

H 2 S: 10.0

N H 3: 25. (3.5 m L con N H 3)

300 ml H 2 O

Started spark at 5:30 p m. Monday March 24, 1958

after a few minutes there was yellowing of the solution but no green

A photograph of a box of his vials labeled as C H 4, N H 3, H 2 O, H 2 S plus spark p 114 Run 22 is shown.

Some of Stanley Miller’s notes from his 1958 hydrogen sulfide (H_2S) experiment along with his original vials.

How the Experiment Was Done According to his laboratory notebook, Miller used the same apparatus as in his original experiment (see Figure 4.2), but the mixture of gaseous reactants included methane (CH_4), carbon dioxide (CO_2), hydrogen sulfide (H_2S), and ammonia (NH_3). After three days of simulated volcanic activity, he collected samples of the liquid, partially purified the chemicals, and sealed the samples in sterile vials. In 2011, Bada’s research team used modern analytical methods to analyze the products in the vials for the presence of amino acids, the building blocks of proteins.

Data from the Experiment The table below shows 4 of the 23 amino acids detected in the 2011 analysis of the samples from Miller’s 1958 H_2S experiment.

A table has 4 rows and 3 columns.

Product compound	Molecule formula	Molar ratio (relative to Glycine)
Glycine	C 2 H 5 N O 2	1
Serine	C 3 H 7 N O 3	3.0 times 10 raised to the negative second power
Methionine	C 5 H 11 N O 2 S	1.8 times 10 raised to the negative third power
Alanine	C 3 H 7 N O 2	1.1

Data from E. T. Parker et al., Primordial synthesis of amines and amino acids in a 1958 Miller H_2S-rich spark discharge experiment, *Proceedings of the National Academy of Sciences USA* 108:5526-5531 (2011). www.pnas.org/cgi/doi/10.1073/pnas.1019191108.

INTERPRET THE DATA

- A *mole* is the number of particles of a substance with a mass equivalent to its molecular (or atomic) mass in daltons. There are 6.02×1023 molecules (or atoms) in 1.0 mole (Avogadro's number; see Concept 3.2). The data table shows the "molar ratios" of some of the products from the Miller H_2S experiment. In a molar ratio, each unitless value is expressed relative to a standard for that experiment. Here, the standard is the number of moles of the amino acid glycine, which is set to a value of 1.0. For instance, serine has a molar ratio of $3.0 \times 10-2$, meaning that for every mole of glycine, there is $3.0 \times 10-2$ mole of serine.
- Give the molar ratio of methionine to glycine and explain what it means.
- How many molecules of glycine are present in 1.0 mole?
- For every 1.0 mole of glycine in the sample, how many molecules of methionine are present? (Recall that to multiply two numbers with exponents, you add their exponents; to divide them, you subtract the exponent in the denominator from that in the numerator.)
-
  - Which amino acid is present in higher amounts than glycine?
- How many more molecules of that amino acid are present than the number of molecules in 1.0 mole of glycine?
- The synthesis of products is limited by the amount of reactants.
- If one mole each of CH_4, NH_3, H_2S, and CO_2 is added to 1 liter of water (=55.5 moles of H_2O) in a flask, how many moles of hydrogen, carbon, oxygen, nitrogen, and sulfur are in the flask?
- Looking at the molecular formula in the table, how many moles of each element would be needed to make 1.0 mole of glycine?
- What is the maximum number of moles of glycine that could be made in that flask, with the specified ingredients, if no other molecules were made? Explain.
- If serine or methionine were made individually, which element(s) would be used up first for each? How much of each product could be made?
- The earlier published experiment carried out by Miller did not include H_2S in the reactants (see Figure 4.2). Which of the compounds shown in the data table can be made in the H_2S experiment

but could not be made in the earlier experiment?

Instructors: A version of this Scientific Skills Exercise can be assigned in **Mastering Biology**.

In Concept 3.2, you learned about evidence for the presence of water on Mars. Even more exciting, in 2018, NASA reported that the rover *Curiosity* had found carbon-based compounds on Mars in a crater where a lake once existed. While these compounds might have been brought to Mars on a meteorite or formed by geologic processes, an intriguing possibility is that they might have been the relics of life-forms that once existed on that planet.

The overall percentages of the major elements of life—C, H, O, N, S, and P—are quite uniform from one organism to another, reflecting the common evolutionary origin of all life. Because of carbon's ability to form four bonds, however, this limited assortment of atomic building blocks can be used to build an inexhaustible variety of organic molecules. Different species of organisms, and different individuals within a species, are distinguished by variations in the types of organic molecules they make. In a sense, the great diversity of living organisms we see on the planet (and in fossil remains) is made possible by the unique chemical versatility of the carbon atom.

Interview

Interview with Stanley Miller: Investigating the origin of life

A photo of Stanley Miller.

Concept Check 4.1

- **VISUAL SKILLS** See Figure 4.2. Miller carried out a control experiment without discharging sparks and found no organic compounds. What might explain this result?

*For suggested answers, see*Appendix A.**

Concept 4.2: Carbon atoms can form diverse molecules by bonding to four other atoms

The key to an atom's chemical characteristics is its electron configuration. This configuration determines the kinds and number of bonds an atom will form with other atoms. Recall that it is the valence electrons, those in the outermost shell, that are available to form bonds with other atoms.

The Formation of Bonds with Carbon

Carbon has 6 electrons, with 2 in the first electron shell and 4 in the second shell; thus, it has 4 valence electrons in a shell that can hold up to 8 electrons. A carbon atom usually completes its valence shell by sharing its 4 electrons with other atoms so that 8 electrons are present. Each pair of shared electrons constitutes a covalent bond (see Figure 2.10d). In organic molecules, carbon usually forms single or double covalent bonds. Each carbon atom acts as an intersection point from which a molecule can branch off in as many as four directions. This enables carbon to form large, complex molecules.

When a carbon atom forms four single covalent bonds, the arrangement of its four hybrid orbitals causes the bonds to angle toward the corners of an imaginary tetrahedron. The bond angles in methane (CH_4) are 109.5°**(**Figure 4.3a**)**, and they are roughly the same in any group of atoms where carbon has four single bonds. For example, ethane (C_2H_6) is shaped like two overlapping tetrahedrons **(Figure 4.3b)**. In molecules with more carbons, every grouping of a carbon bonded to four other atoms has a tetrahedral shape. But when two carbon atoms are joined by a double bond, as in ethene (C_2H_4), the bonds from both carbons are all in the same plane, so the atoms joined to those carbons are in the same plane as well **(Figure 4.3c)**. We find it convenient to write molecules as structural formulas, as if the molecules being represented are two-dimensional, but keep in mind that molecules are three-dimensional and that the shape of a molecule is central to its function.

Figure 4.3

The shapes of three simple organic molecules.

A table has 3 rows and 5 columns.

The rows are under the column for Shape, and they are labeled a through c.

- (a) Methane, C H 4, is a tetrahedral molecule. The four hydrogen atoms of methane are drawn as the four corners of a triangular pyramid, with the carbon atom in the center.
- (b) Ethane, C 2 H 6, is a molecule with all single bonds and more than one tetrahedral carbon. It has a distinctive shape, as though two tripods were attached to each other with all six legs pointing out.
- (c) Ethene (ethylene), C 2 H 4, contains a double bond between two carbons. It has the shape of two Y's attached bottom-to-bottom. All the atoms connected to a double bond are in the same plane, so this molecule is flat.

The number of electrons required to fill the valence shell of an atom is generally equal to the atom's **valence**, the number of covalent bonds it can form. **Figure 4.4** shows the valences of carbon and its most frequent bonding partners—hydrogen, oxygen, and nitrogen. These are the four main atoms in organic molecules.

Figure 4.4

Valences of the major elements of organic molecules.

Valences of the major elements of organic molecules.

Valence, the number of covalent bonds an atom can form, is generally equal to the number of electrons required to fill the valence shell. (Sodium, phosphorus, and chlorine are exceptions.)

Hydrogen (valence = 1). Electron distribution diagram with a single electron in the valence shell, and a Lewis dot structure showing an H with one dot.

Oxygen (valence = 2). Electron distribution diagram: two electrons in the first shell and six electrons in the second shell. Lewis dot structure: O surrounded by 6 dots.

Nitrogen (valence = 3). Electron distribution diagram: two electrons in the first shell and five electrons in the second shell. Lewis dot diagram: N surrounded by 5 dots.

Carbon (valence = 4). Electron distribution diagram: two electrons in the first shell and four electrons in the second shell. Lewis dot diagram: C surrounded by four dots.

MAKE CONNECTIONS *Draw the Lewis dot structures for sodium, silicon, phosphorus, sulfur, and chlorine. (Refer to*Figure 2.7.*)*

*For suggested answer, see*Appendix A.*

The electron configuration of carbon gives it covalent compatibility with many different elements. Let's consider how valence and the rules of covalent bonding apply to carbon atoms with partners other than hydrogen. We'll look at two examples, the simple molecules carbon dioxide and urea.

In the carbon dioxide molecule (CO_2), a single carbon atom is joined to two atoms of oxygen by double covalent bonds. The structural formula for CO_2 is shown here:

O=C=O

Each line in a structural formula represents a pair of shared electrons. Thus, the two double bonds in CO_2 have the same number of shared electrons as four single bonds. The arrangement completes the valence shells of all atoms in the molecule:

C O 2 with electron shells.

C has 2 shells. Its inner shell has 2 electrons. Its outer shell has 4 electrons is shares with O on its left, and 4 electrons it shares with O on its right. Each O has 2 shells. Each inner shell has 2 electrons, and each outer shell has 4 electrons plus the 4 shared electrons with C.

Because CO_2 is a very simple molecule and lacks hydrogen, it is often considered inorganic, even though it contains carbon. Whether we call CO_2 organic or inorganic, however, it is clearly important to the living world as the source of carbon, via photosynthetic organisms, for all organic molecules in organisms (see Concept 2.4).

Urea, $CO(NH_2)_2$, is an organic compound found in urine. Again, each atom has the required number of covalent bonds. In this case, one carbon atom participates in both single and double bonds.

Structural formula of urea. The carbon atom is double bonded to an oxygen atom, and forms single bonds with two different nitrogen atoms, forming a total of four bonds.

Urea and carbon dioxide are molecules with only one carbon atom. But a carbon atom can also use one or more valence electrons to form covalent bonds to other carbon atoms, linking the atoms into chains, as shown here for C_3H_8:

Shells and electrons for C 3 H 8.

Three C atoms are in a row, each with 2 shells. Each inner shell has 2 electrons. The outer shells share 2 electrons with the adjacent C. The outer C atoms each have 3 H atoms, and the inner C has 2 H atoms. The outer shell of each C have 2 electrons they share with the single shell of each H.

Molecular Diversity Arising from Variation in Carbon Skeletons

Carbon chains form the basis of most organic molecules. Carbon skeletons vary in length and may be straight, branched, or arranged in closed rings **(Figure 4.5)**. Some carbon chains have double bonds, which vary in number and location. Such variation in carbon chains is one important source of the molecular complexity and diversity that characterize living matter. In addition, the skeletons of biological molecules often include atoms of other elements, like oxygen and phosphorus; such atoms can also be bonded to carbons of the skeleton.

Figure 4.5

Four ways that carbon skeletons can vary.

Four ways that carbon skeletons can vary: length, branching, double bond position, and presence of rings.

The four ways are as follows.

- a. Carbon skeletons vary in length. Ethane (with two carbons) and propane (with three carbons) have carbon skeletons with different lengths.
- b. Skeletons may be unbranched or branched. Butane has four carbons in a row and no branching. Isobutane (2-methylpropane), has three carbons in a row, but the middle carbon also has another

carbon branching off.

- c. The skeleton may have double bonds, which can vary in location. 1-butene has a double bond between the first two carbon atoms; 2-butene has a double bond between the second and third carbon atoms.
- d. Some carbon skeletons are arranged in rings. In the abbreviated structural formula for each compound (to its right), each corner represents a carbon and its attached hydrogens. Cyclohexane (C_6H_{12}) and benzene (C_6H_6) are examples of molecules with rings of six carbon atoms. Benzene is drawn with alternating double and single bonds in the ring, whereas cyclohexane contains all single bonds.

Animation

Diversity of Carbon-Based Molecules

image

Play

image

Play

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Hydrocarbons

All of the molecules that are shown in Figures 4.3 and 4.5 are **hydrocarbons**, organic molecules consisting of only carbon and hydrogen. Atoms of hydrogen are attached to the carbon skeleton wherever electrons are available for covalent bonding. Hydrocarbons are the major components of petroleum, which is called a fossil fuel because it consists of the partially decomposed remains of organisms that lived millions of years ago.

Interview

Interview with Deborah Gordon: Studying How Ants Use Hydrocarbons to Communicate

A photo of Deborah Gordon.

Although hydrocarbons are not prevalent in most living organisms, some of a cell's organic molecules have regions consisting of only carbon and hydrogen. For example, the molecules known as fats have long hydrocarbon tails attached to a nonhydrocarbon component **(Figure 4.6)**. Neither petroleum nor fat dissolves in water; both are hydrophobic compounds because the great majority of their bonds are relatively

nonpolar carbon-to-hydrogen linkages. Another characteristic of hydrocarbons is that they can undergo reactions that release a relatively large amount of energy. The gasoline that fuels a car consists of hydrocarbons, and the hydrocarbon tails of fats serve as stored fuel for plant embryos (seeds) and animals.

Figure 4.6

The role of hydrocarbons in fats.

Part of a human adipose cell. Fat droplets with approximately 10 micrometer diameters clustered in adipose tissue.

(a) Mammalian adipose cells stockpile fat molecules as a fuel reserve. This colorized micrograph shows part of a human adipose cell with many fat droplets, each containing a large number of fat molecules. **(b)** A fat molecule consists of a small, nonhydrocarbon component joined to three hydrocarbon tails that account for the hydrophobic behavior of fats. The tails can be broken down to provide energy.
(Black = carbon; gray = hydrogen; red = oxygen.)

MAKE CONNECTIONS *How do the tails account for the hydrophobic nature of fats? (See*Concept 3.2.)**

*For suggested answer, see*Appendix A.**

Isomers

Variation in the architecture of organic molecules can be seen in **isomers**, compounds that have the same numbers of atoms of the same elements but different structures and hence different properties. We will examine three types of isomers: structural isomers, *cis-trans* isomers, and enantiomers.

Structural isomers differ in the covalent arrangements of their atoms. Compare, for example, the two five-carbon compounds in **Figure 4.7a**. Both have the molecular formula C_5H_{12}, but they differ in the covalent arrangement of their carbon skeletons. The skeleton is straight in one compound but branched in the other. The number of possible isomers increases tremendously as carbon skeletons increase in size. There are only three forms of C_5H_{12} (two of which are shown in Figure 4.7a), but there are 18 variants of C_8H_{18} and 366,319 possible structural isomers of $C_{20}H_{42}$. Structural isomers may also differ in the location of double bonds.

Figure 4.7

Three types of isomers.

Figure C is for enantiomers.

Isomers are compounds that have the same molecular formula but different structures.

Enantiomers differ in spatial arrangement around an asymmetric carbon, resulting in molecules that are mirror images, like left and right hands. The two isomers here are designated L and D isomers from the Latin for left and right (levo and dextro). Enantiomers cannot be superimposed on each other. The L isomer has a C atom at the center of a tetrahedron with C O 2 H above, H down and left, N H 2 down and right, and C H 3 down and toward the page. The D isomer switches the positions of N H 2 and H.

DRAW IT *There are three structural isomers of*C5H12*;* *draw the one not shown in (a).*

*For suggested answer, see*Appendix A.*

Animation

Isomers

image

Play

image

Play

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In ***cis-trans*isomers** (also known as *geometric isomers*), carbons have covalent bonds to the same atoms, but these atoms differ in their spatial arrangements due to the inflexibility of double bonds. Single bonds allow the atoms they join to rotate freely about the bond axis without changing the compound. In contrast, double bonds do not permit such rotation. If a double bond joins two carbon atoms, and each C also has two different atoms (or groups of atoms) attached to it, then two distinct *cis-trans* isomers are possible. Consider a simple molecule with two double-bonded carbons, each of which has an H and an X attached to it **(Figure 4.7b)**. The arrangement with both Xs on the same side of the double bond is called a *cis isomer,* and that with the Xs on opposite sides is called a *trans isomer*. The subtle difference in shape between such isomers can have a dramatic effect on the biological activities of organic molecules. For example, the biochemistry of vision involves a light-induced change of retinal, a chemical compound in the eye, from the *cis* isomer to the *trans* isomer (see Figure 50.17). Another example involves *trans* fats, harmful fats formed during food processing that are discussed in Concept 5.3.

Enantiomers are isomers that are mirror images of each other and that differ in shape due to the presence of an *asymmetric carbon*, one that is attached to four different atoms or groups of atoms. (See the middle carbon in the ball-and-stick models shown in **Figure 4.7c**.) The four groups can be arranged in space around the asymmetric carbon in two different ways that are mirror images. Enantiomers are, in a way, left-handed and right-handed versions of the molecule. Just as your right hand won't fit into a left-handed glove,

a “right-handed” molecule won’t fit into the same space as the “left-handed” version. Usually, only one isomer is biologically active because only that form can bind to specific molecules in an organism.

The concept of enantiomers is important in the pharmaceutical industry because the two enantiomers of a drug may not be equally effective, as is the case for both ibuprofen and the asthma medication albuterol **(Figure 4.8)**. Methamphetamine also occurs in two enantiomers that have very different effects. One enantiomer is the highly addictive stimulant drug known as “crank,” sold illegally in the street drug trade. The other has a much weaker effect and is the active ingredient in an over-the-counter vapor inhaler for treatment of nasal congestion. The differing effects of enantiomers in the body demonstrate that organisms are sensitive to even the subtlest variations in molecular architecture. Once again, we see that molecules have emergent properties that depend on the specific arrangement of their atoms.

Figure 4.8

The pharmacological importance of enantiomers.

Two enantiomers for the drug ibuprofen and two enantiomers for the drug Albuterol.

The S-ibuprofen and R-ibuprofen molecules are mirror images of each other, and only the S-enantiomer is effective at reducing inflammation and pain. The R and S enantiomers are mirror images. Only R-Albuterol is effective at relaxing bronchial (airway) muscles and improving airflow in asthma patients.

Ibuprofen and albuterol are drugs whose enantiomers have different effects. (S and R are used here to distinguish between enantiomers, rather than D and L as in Figure 4.7c.) Ibuprofen is commonly sold as a mixture of the two enantiomers; the S enantiomer is 100 times more effective than the R form. Albuterol is synthesized and sold only as the R form of that drug; the S form counteracts the active R form.

Concept Check 4.2

- **DRAW IT** (a) Draw a structural formula for C_2H_4. (b) Draw the *trans* isomer of $C_2H_2Cl_2$.
- **VISUAL SKILLS** Which two pairs of molecules in Figure 4.5 are isomers? For each pair, identify the type of isomer.
- How are gasoline and fat chemically similar?
- **VISUAL SKILLS** See Figures 4.5a and 4.7. Can propane (C_3H_8) form isomers? Explain.

*For suggested answers, see*Appendix A.**

Concept 4.3: A few chemical groups are key to molecular function

The distinctive properties of an organic molecule depend not only on the arrangement of its mostly carbon skeleton but also on the various chemical groups attached to that skeleton. These groups may participate in chemical reactions or may contribute to function indirectly by their effects on molecular shape; they help give each molecule its unique properties.

The Chemical Groups Most Important in the Processes of Life

Consider the differences between estradiol (a type of estrogen) and testosterone. These compounds are female and male sex hormones, respectively, in humans and other vertebrates. Both are steroids, organic molecules with a common carbon skeleton in the form of four fused rings. They differ only in the chemical groups attached to the rings (shown here in abbreviated form, where each corner represents a carbon and its attached hydrogens); the distinctions in molecular architecture are shaded in blue:

Structural differences between the hormones estradiol and testosterone.

In estradiol, one of four fused rings is aromatic, containing three double bonds, while in testosterone, this ring contains only one double bond. Estradiol has an alcohol (O H) where testosterone has a ketone, and testosterone has a methyl (C H 3) group that is not present in estradiol.

The different actions of these two molecules on many targets throughout the body are the basis of sexual characteristics, producing the contrasting features of male and female vertebrates. In this case, the chemical groups are important because they affect molecular shape, contributing to function.

In other cases, chemical groups are directly involved in chemical reactions; such groups are known as **functional groups**. Each has certain properties, such as shape and charge, that cause it to participate in chemical reactions in a characteristic way.

The seven chemical groups most important in biological processes are the **hydroxyl group**, **carbonyl group**, **carboxyl group**, **amino group**, **sulfhydryl group**, **phosphate group**, and **methyl group**. The first six groups can be chemically reactive; of these six, all except the sulfhydryl group are also hydrophilic and thus increase the solubility of organic compounds in water. The methyl group is not reactive, but instead often serves as a recognizable tag on biological molecules. Study **Figure 4.9** to become familiar with these biologically important chemical groups. Note in the Examples column of the figure that the carboxyl group and the amino group are ionized at normal cellular pH.

Figure 4.9

Some biologically important chemical groups.

A table has 7 rows and 3 columns.

The columns are chemical group, group properties and compound names, and examples. Row entries are as follows.

Chemical group: Hydroxyl group (single bond, O H). The group is polar and can form hydrogen bonds. Compound name: Alcohol. Hydroxyl groups have the structure single bond, O H. The group is polar and can form hydrogen bonds (for example, with water, helping dissolve compounds such as sugars). Sugars have many hydroxyl groups. Compounds that contain an hydroxyl group are also called alcohols. An example is ethanol, the alcohol in alcoholic beverages.

Chemical group: Carbonyl group, which has the structure C O, where a carbon and oxygen are double bonded. Compound name: ketone or aldehyde. Carbonyl groups have the structure C O, where a carbon and oxygen are double bonded. Carbonyl groups at the end of a carbon chain are aldehydes (these also have a hydrogen on the carbonyl carbon). Carbonyl groups in the middle of a carbon chain are ketones. Propanol, with three carbons, is an aldehyde. Acetone, also with three carbons, is the simplest ketone. A sugar containing an aldehyde group is called an aldose, and a sugar containing a ketone group is called a ketose.

Chemical group: Carboxyl group (structure C O O H). Acts as acid (can donate H plus) because the covalent bond between oxygen and hydrogen is so polar. Compound name: Carboxylic acid or organic acid. Carboxyl groups have the structure single bond, C O O H. The carbon has a double bond to one oxygen and a single bond to the O H. Carboxyl groups act as acids, ionizing to form carboxylate ion, single bond, C O O minus and H plus. A carboxylate ion is the same as a carboxyl group, but has a negative charge on the oxygen that lost a hydrogen. Compounds containing the carboxyl group are known as carboxylic acids or organic acids. An example is acetic acid, a two-carbon acid that gives vinegar its sour taste. Ionized form of single bond, C O O H (carboxylate ion), is found in cells.

Chemical group: Sulfhydryl group (single bond, S H). Compound name: Thiol. Sulfhydryl groups have the structure single bond, S H. Two sulfhydryl groups can react, forming a disulfide cross-link of the form single bond, S, single bond, S, single bond, that helps stabilize protein structure. Hair protein cross-links maintain the straightness or curliness of hair. In hair salons, these crosslinks are broken and re-formed during a permanent treatment. Compounds containing sulfhydryl groups are known as thiols. An example is the amino acid cysteine, which contains a sulfhydryl group at the end of its side-chain.

Chemical group: Phosphate group (single bond, O P O 3 superscript minus 2). Contributes negative charge (1 minus when position inside a change of phosphates; 2 minus when at the end). Compound name: Organic phosphate. Phosphate groups consist of a phosphorus atom surrounded by four oxygen atoms. They can

bond to a carbon or other atom through one or two of their four oxygen atoms. The structure can be written as single bond, O P O 3 superscript minus 2. They have a charge of minus 2 when positioned at the end of a chain, or minus 1 when in the middle of a chain. When an attached phosphate group is hydrolyzed (broken off) by water, energy is released. Carbon compounds containing phosphate are called organic phosphates. An example is glycerol phosphate, a three-carbon compound with a phosphate on carbon 1, and hydroxy groups on carbons 2 and 3.

Chemical group: Methyl group (single bond, C H 3). Affects the expression of genes when on DNA or on proteins bound to DNA. Compound name: Methylated compound. Methyl groups have the structure single bond, C H 3, and are ubiquitous in organic chemistry. When present in D N A or proteins bound to D N A, they can affect gene expression. The presence of methyl groups also affects the shape and function of male and female sex hormones. Compounds containing methyl groups can be called methylated compounds. An example is 5-methyl cytosine, the methylated form of the D N A base cytosine.

Animation

Functional Groups

image

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ATP: An Important Source of Energy for Cellular Processes

The “Phosphate group” row in Figure 4.9 shows a simple example of an organic phosphate molecule. A more complicated organic phosphate, **adenosine triphosphate**, or **ATP**, is worth mentioning here because its function in the cell is so important. ATP consists of an organic molecule called adenosine attached to a string of three phosphate groups:

Structure of A T P.

Adenosine attached to a tail of three phosphate groups. Each phosphate group consists of a central P that forms 5 bonds: a double bond to one O and three single covalent bonds to three other oxygen atoms. Oxygen atoms forming only single covalent bonds in this molecule are shown with negative charges.

Where three phosphates are present in series, as in ATP, one phosphate may be split off as a result of a reaction with water. This inorganic phosphate ion, $HOPO_3^{2-}$, is often abbreviated Ⓟi in this book, and a phosphate group in an organic molecule is often written as Ⓟ. Having lost one phosphate, ATP becomes adenosine *di*phosphate, or ADP. Although ATP is sometimes said to store energy, it is more accurate to think of it as storing the potential to react with water or other molecules. Overall, the process releases energy that can be used by the cell. You'll learn more about this in Concept 8.3.

A T P reacts with water to produce inorganic phosphate, A D P and energy.

The reactant A T P (an adenosine attached to 3 P's representing phosphate groups) reacts with water, producing P sub i (inorganic phosphate), A D P (adenosine attached to 2 P's) and energy.

The Chemical Elements of Life: *A Review*

Living matter, as you have learned, consists mainly of carbon, oxygen, hydrogen, and nitrogen, with smaller amounts of sulfur and phosphorus. These elements all form strong covalent bonds, an essential characteristic in the architecture of complex organic molecules. Of all these elements, carbon is the virtuoso of the covalent bond. The versatility of carbon makes possible the great diversity of organic molecules, each with particular properties that emerge from the unique arrangement of its mostly carbon skeleton and the chemical groups attached to that skeleton. This variation at the molecular level provides the foundation for the rich biological diversity found on our planet.

Concept Check 4.3

- **VISUAL SKILLS** What does the term *amino acid* signify about the structure of such a molecule? See Figure 4.9.
- What chemical change occurs to ATP when it reacts with water and releases energy?
- **DRAW IT** Suppose you had an organic molecule such as cysteine (see Figure 4.9, sulfhydryl group example), and you chemically removed the —NH_2 group and replaced it with —COOH. Draw this structure. How would this change the chemical properties of the molecule? Is the central carbon asymmetric before the change? After?

*For suggested answers, see*Appendix A.**